INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

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(Affiliated to UEM Kolkata and Approved by AICTE)

Degree: BCA, Stream: BCA

Term – I , Third Semester Examination, September - 2024

Subject Code:BCAMD301

Subject Name: Computer Networking

Full Marks:30 Duration: 1 Hour

Part - A

Attempt 5 questions

Each question carries 2 marks (2 × 5)

1. What is the concept of classful addressing? 2

   or

   What is the address space of IPV4? 2

2. Find the error, if any, in the following IP address: 2
   75.45.301.14

   or

   Find the class of the address: 2
   11000001 10000011 00011011 11111111

3. Change the following IP address from binary notation to dotted-decimal notation. 2
   10000001 00001011 00001011 11101111

   or

   Change the following IP addresses from binary notation to hexadecimal notation. 2
   10000001 00001011 00001011 11101111

4. What is Half Duplex? 2

   or

   What is Multipoint connection? 2

5. Define bit rate with an example 2

   or

   Define bandwidth with an example 2

Part - B

Attempt 2 questions

Each question carries 5 marks (5 × 2)

6. If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700, and 900 Hz, what is its bandwidth? Draw the spectrum, assuming all components have a maximum amplitude of 10 V. 5

or

Demonstrate Shannon Capacity. 5

7. Explain throughput of a network with an example. 5

or

Explain latency (delay) of a network with an example. 5

Part - C

Attempt 1 question

Each question carries 10 marks (10 × 1)

8. Describe peer to peer communication with respect to data transfer using reference model. 10

or

Describe the functionality of each layer of the Internet Model. 10